

Challenges and Opportunities for Playful Technology in Health Prevention: Using Virtual Reality to Supplement Breastfeeding Education

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ABSTRACT

Playful technology offers the opportunity to engage users, convey knowledge and prompt reflection. We built on this potential and designed a VR simulation to give parents-to-be insights into the lived breastfeeding experience. An evaluation with 10 participants revealed that users appreciated the system but perceived similarity between the simulation and games, leading to conflicting expectations. Reflecting on this, we outline challenges for playful VR simulation design for healthcare contexts.

Index Terms: Human-centered computing—Human computer interaction (HCI); Human-centered computing—Virtual reality

1 INTRODUCTION AND BACKGROUND

Breastfeeding has been associated with many health benefits for both parents and children [22]. However, the feeding process is often deemed as challenging as many new parents struggle to establish a positive feeding relationship and are surprised by the challenges, particularly in the early postnatal stage [10, 20]. Many parents reported having to bare with uncertainties throughout the feeding journey and insecurity about their ability to breastfeeding in addition to experiencing pain and physical discomfort [6, 15, 22]. Breastfeeding education seeks to address these challenges through the provision of knowledge (e.g., on how to recognise the needs of a baby or how to latch a baby) [7]. However, prior work [7, 17] suggests that antenatal education does not adequately reflect the complexity of the lived experience of breastfeeding [7, 20]. As a results, new parents are struck by the challenges, leaving them in a vulnerable position [20].

Playful technology in general, and Virtual Reality (VR) in particular has the potential to support health prevention: Digital games that directly invite users to engage in interaction are an appealing approach to work with a wide range of audiences; games have been effectively applied in many health-related settings, for example, to encourage healthy eating [16], to promote physical activity [14], or to provide social support for patients [8]. This concept leverages game elements such as points, badges and leaderboards, which provide clear goals and feedback, hence, encouraging users to complete tasks and improve their performance [18]. Likewise, VR can be designed to give users a glimpse of otherwise inaccessible experiences, presenting them from an immersive first-person view, while being suited for the conveyance of challenging experiences (e.g., see [5] and [3]). Therefore, we assume that VR offers a unique opportunity to allow parents-to-be to experience breastfeeding in different settings, providing them with insights into the feeding process and the lived breastfeeding experience [10].

In this paper, we explore opportunities and challenges that we encountered throughout design and evaluation of VirtualFeed [20], a VR breastfeeding simulation that offers parents-to-be an insight

into the lived experience of breastfeeding. In particular, we focus on the implications that similarities and differences between games and our simulation had for the way that users approached the systems, and for the expectations that were raised. We report findings from a qualitative user study with ten participants that highlighted the tensions that emerge when users engage with a simulation, but would expect game elements to guide them through their experience. On this basis, we discuss advantages and disadvantages when leveraging playful technology in the context of health prevention, focusing on how the presence or absence of game elements can impact user experience.

2 VIRTUALFEED: A VR BREASTFEEDING SIMULATION

In our previous work [19], we developed *VirtualFeed*, a VR simulation that enables users to experience breastfeeding in different settings. We co-designed the simulation with breastfeeding parents, parents-to-be and their partners with a focus on aspects such as the uncertainty while feeding a newborn baby and the influence of the environment on the feeding experience. Here, we give an overview of the simulation with respect to technical setup, simulated behaviour of a newborn baby, simulated breastfeeding experience and breastfeeding settings.

2.1 Technical Setup

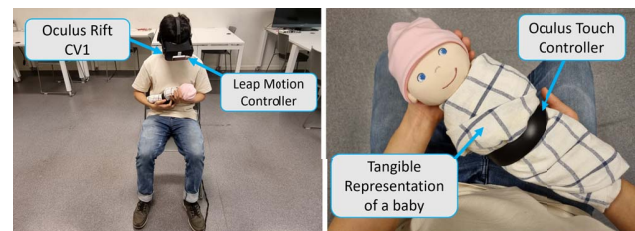


Figure 1: Simulation setup from the third person and the user perspective

Our simulation combines an Oculus Rift CV1, a tangible probe representing a baby, and a Leap Motion Controller (see figure 1). Regarding the setup of the hardware, we attach an Oculus Touch controller to the tangible representation of a baby so that we can track the pose of the baby and map it to a virtual presentation of a baby in VR space. We mount the Leap Motion Controller to the front of the VR headset to enable hand interactions with the virtual baby in VR. With respect to the software platform, our simulation was developed on Unity using SDKs from Oculus and Leap Motion. Within the simulated experience, we provide a virtual self-avatar representing the user in a sitting pose. Here, we map the hand interactions captured by the Leap Motion controller and the pose of the user's head to the rig of the avatar. For upper body parts that are not tracked by hardware or when the user's hands are occluded by the tangible baby, we estimate the pose of those parts using the Forward And Backward Reaching Inverse Kinematics (FABRIK) [2] and map the estimated pose to the avatar rig.

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2.2 Simulated Behaviour of a Newborn Baby

To convey the uncertainty of feeding a newborn baby, we provide a rigged avatar representing a baby, which we programmed and animated to react to users interactions (e.g., latching to the nipple) or reprogrammed events in the virtual space. The virtual baby communicates with the user through small sounds and non-verbal cues such as limbs movement or facial expressions. We modelled the behaviour of the baby based on a state machine diagram (see figure 2) consisting of 9 states, each can be triggered by user inputs or events in virtual environment.

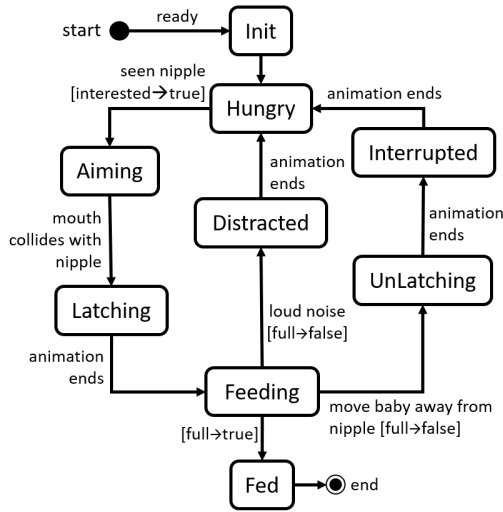


Figure 2: state-machine-diagram modelling the behaviours of the virtual baby

2.3 Simulated Breastfeeding Experience

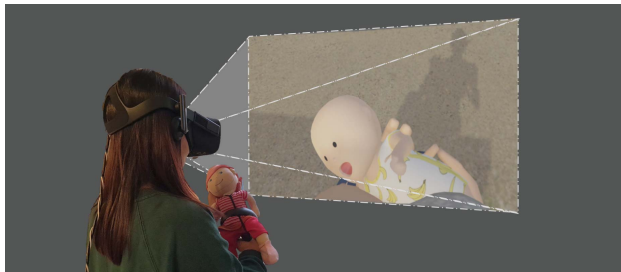


Figure 3: A person interacting with the simulation with a screenshot rendered from the user's view

We simulate a common breastfeeding experience that allows the user to try latching the virtual baby and breastfeed until the baby is full by mean of moving the tangible probe representing the baby closer to or away from their chest. Within the VR space, the user can see a virtual representation of themselves in forms of an avatar and they can aim the virtual baby's mouth close to their virtual breast to initiate the latching process. Once the baby is latched, the user needs to maintain the feed until the virtual baby is full and gently adjust their sitting position to ensure their comfort while not interrupting the baby. There are only sparse cues provided to the user that mimic those of real-world breastfeeding (e.g., the baby making cues when hungry, and unlatching when full). We intentionally decided not to include further elements that would highlight this process to users

in an effort to maintain the uncertainty of real-world breastfeeding within the simulation.

2.4 Breastfeeding Settings

Our simulation consists of three breastfeeding settings, at one's home, in a public park, and in a meeting room. These settings hint on the impact of the environment on the feeding experience. In each setting, we include animated non-player characters with the intention to encourage the user to reflect on breastfeeding experiences with presence of other people.



Figure 4: Simulation setup from the third person and the user perspective

3 USER STUDY

We carried out a comprehensive qualitative evaluation of VirtualFeed that included extensive engagement with the simulation in combination with interviews. In this paper, we focus on the subset of results that addresses the general potential of the simulation to convey the lived experience of breastfeeding, and aspects of interactivity and the inclusion of playful elements in our system.

3.1 Participants and Procedure

We recruited 10 participants (9 females) through a snowball sampling method among friends and colleagues and with recruitment flyers distributed at a local day care centre, a university campus and social media. Participants are in 26-35 age group (N=8) and 36-45 age group (N=2). Participants are breastfeeding parents (N=6), parents-to-be (N=3) and partner (N=1). All participants reside in Western Europe. We invited the participants to take part individually in an one hour study. Each study session consisted of two parts: experiencing the VR breastfeeding simulation and a semi-structured interview. Participants gave informed consent on an online form before the study took place and were asked to give consent verbally at the beginning of the study. Then, they participated in interviews about their perspective toward the simulation, where we asked questions such as "What is your first impression of the simulation?" and "How was the experience of feeding the baby for you?". At end of the session, Participants were given the opportunity to ask questions and were compensated with a 20 Euro cash for their participation. All sessions were conducted in English, audio-recorded and then transcribed for analysis. The research protocol was approved by the institutional ethics committee.

3.2 Data Analysis

We followed a deductive thematic analysis approach [4] to examine interview transcripts with respect to the potential of VR as a tool to convey the lived experience, and the role of interactivity and playful elements. The analysis was led by the first author who read the transcript multiple times to achieve familiarisation, coded and aggregated instances from the transcript.

3.3 Results

Here, we present a subset of our interview findings with respect to *VR as a Tool to Convey the Lived Experience of Breastfeeding and the Role of Interactivity and Playful Elements*.

3.3.1 VR as a Tool to Convey the Lived Experience of Breastfeeding

Our data suggests that participants were generally enthusiastic about the simulation, for example, they described their experience as *"fun"* (P3), *"impressive"* (P8), and *"playful"* (P5), already hinting at the similarity between their experience with the simulation and a game. With respect to realism, participants pointed out that the simulated experience is reminiscent of the early days of breastfeeding experience in a way that *"it brought nice memories, from the early months. I think it's umm, helps to recreate the experience of early month breastfeeding."* Participants noted that the simulation realistically conveyed nuances and the uncertainty within the feeding process; one breastfeeding parent commented *"the baby takes time too latch so that was really realistic, so give you the expectation that it's not, as easy that you had to try a bit"* (P4), which provoked frustration manifested through comments like *"It's a bit more stressful like, come on, start."* (P5). With respect to the application of the simulation, participants appreciated the potential of the simulation of being a companion to antenatal education. For instance, it can be used *"as a conversation starter between you and your doctors or just when you are in a group of mothers and they can do this. it is something fun to do you can relieve the pressure a little bit ... that what does it make you feel, what made you think."* (P5). Instances here support the notion that the simulation can give the users a glimpse at the experience of breastfeeding, and encourages them to reflect on the challenges within the lived experience of breastfeeding parents.

3.3.2 The Role of Interactivity and Playful Elements

Throughout our data, there are numerous instances that relate to interactivity and playful elements as an integral element of user experience, with the absence of game elements leading to negative participant responses. Here, we observed that participants anticipated the inclusion of game elements as they commonly associated these with the use of VR systems (despite none of the users being avid VR users). While there was a general consensus among participants that the nuances and uncertainty conveyed in the simulated breastfeeding experience reflect a realistic challenge of breastfeeding, participants noted that the vague reaction of the virtual baby hindered their user experience, and recommended including *"pop up messages like your baby has latched, congratulations!"* (P2) which would make implicit processes explicit, and which is similar to messaging used in games that are meant to alert the player to important achievements. Likewise, to flatten the learning curve, participants suggested that *"more information. Hmm, like a tutorial"* (P3) would be useful, again resembling the experience that players are offered in games. Additionally, participants also highlighted the potential of *"gamification"* (P6) so that the simulation would provide direct feedback on user performance, transforming the simulation into a gamified training tool much more similar to traditional games that provide players with continuous feedback on performance and progress. Finally, we observed instances where participants asked for more context within the simulation, e.g., *"I was surprised to see that in the house the person, so it would be nice if you give the*

context to it." (P6). This suggests that there is room to leverage narrative structures similar to those used in games to embed user experiences in a broader context, for example, by introducing a story line, a common technique used in game design.

4 DISCUSSION

While the results of the interviews show that participants were generally enthusiastic about the breastfeeding simulation, and suggested that the simulated experience gave valuable insights into the lived experience of breastfeeding, the tension that emerged from users hoping to engage with a game rather than a simulation warrants further discussion.

4.1 Tensions Between Simulations and Games

In our work, we originally intended to address the complexity of the lived experience of breastfeeding through a design of a VR simulation that conveys nuances and centers the uncertainty within the breastfeeding process. While our data suggests that the simulated experience succeeded in recreating challenges common to early breastfeeding, the element of uncertainty also impacted the experience that users had. There was a strong anticipation of a game-like simulation, in which user actions are quantified and explicitly reacted upon. This raises a tension between our need to present a realistic yet nuanced experience, and the provision of an experience to users that offers sufficient feedback and scaffolding that is typically provided by games [21]. In our case, we argue that the introduction of such playful elements needs to be subtle as to not distract the user from the actual feeding experience [9], and to avoid contradicting the fact that there are uncertainties throughout the feeding journey that parents need to embrace [13].

4.2 Introducing Playful Elements in Simulations

Our findings have implications for the design of VR simulations beyond our narrow setting of antenatal education and breastfeeding. Considering the widespread popularity of games, many users will bring expectations from games to the simulations they engage with. Most importantly, many games offer users certainty and continuous progress/performance feedback, which is indeed considered a marker of quality (e.g., see [1]), hence users may respond less positively to uncertainty in simulations. Here, contextualizing simulations can be one avenue to prime users to a slightly different experience. Likewise, we want to explore if explicit feedback can be gradually removed as users progress through the simulation. Finally, we speculate that juiciness (i.e., a more implicit form of gamification that is achieved through abundant audiovisual effects [11, 12]) can support feedback provision without making the being too explicit about the user's actions. However, one main challenge for designers would be to ensure that the audiovisual effects are not overwhelming to avoid distracting users from other elements of the simulation they should be focusing on. Here, we see an avenue for future work where we want to extend our simulation to include traditional elements of gamification (e.g., points and explicit feedback) and juiciness, allowing us to compare how it affects user engagement with the simulation, and insights that users gain.

5 CONCLUSION

VR simulations offer the opportunity of making inaccessible experiences - as for example a future breastfeeding experience - accessible in the context of health prevention, allowing users to enact future scenarios, thereby exploring and refining their own perspectives. However, because many prospective users are already familiar with games, it becomes relevant to consider how expectations that are based on prior play experiences affect how users engage with healthcare simulations. Our paper offers a first exploration of the issue, and provides avenues for future work that explores how playful

elements can be integrated in VR simulations for health prevention in a nuanced way.

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